UC Berkeley School of Information

MIMS FINAL PROJECT · 2026

CivicSim

Demographically Grounded Large Language Model (LLM) Simulation of Public Opinion

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01 · BACKGROUND

Who uses population models?

Policymakers, election forecasters, and researchers rely on statistical approximations to predict how populations will respond — before any vote is cast or dollar is spent. These models do not always produce accurate results.

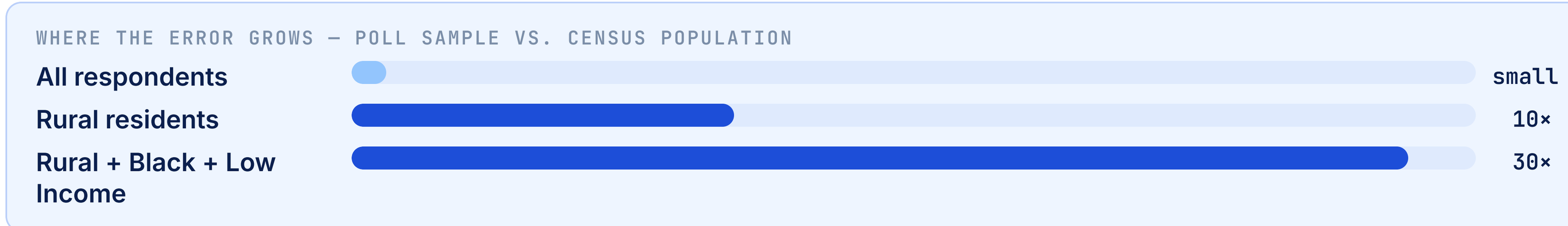
> **2024 Elections**
>
> Standard polls failed to capture a major Hispanic voting shift because these communities were chronically under-sampled.

02 · EXPERIMENTAL FINDINGS

Where do current models fall short?

① Polls survey from the wrong neighborhoods

30× Error rate in representation of rural Black Americans across different Regions.



② Current models primarily focus on marginals, not intersections

`age` or `income` or `location` or `race`

Current research focuses on opinions of marginals, e.g., "What do Black individuals think?" They rarely focus on demographic intersections, "What do young, low income, Black, rural voters from the Midwest think?"

03 · OUR SOLUTION

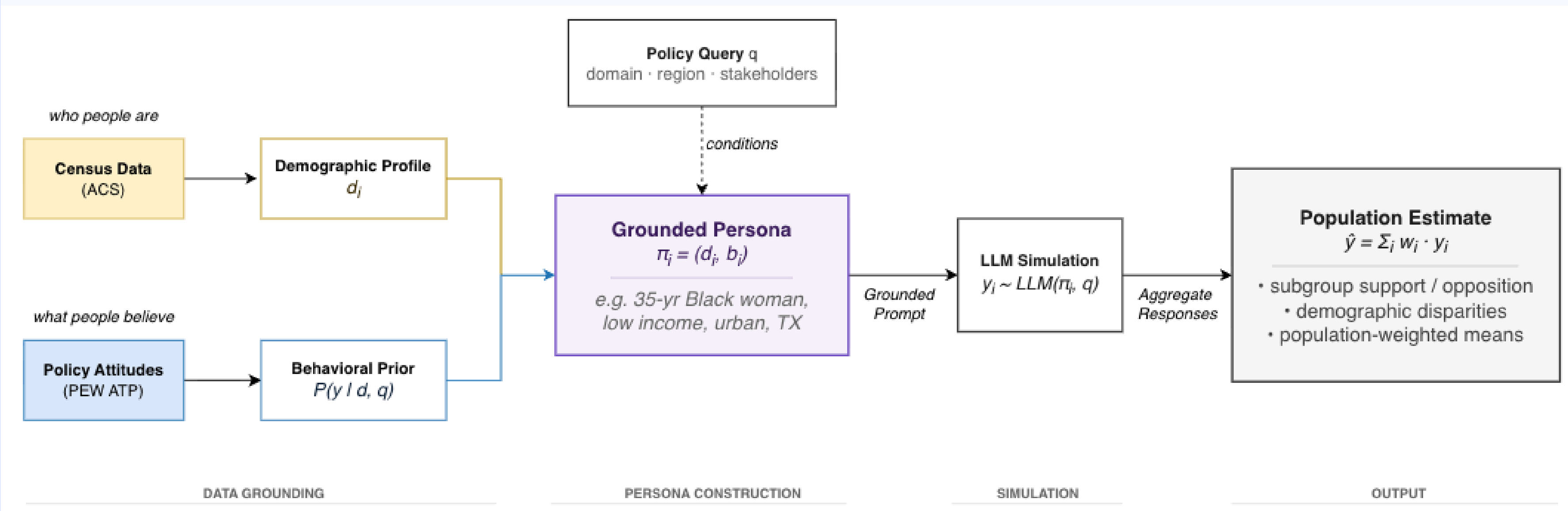


Fig. 1: CivicSim pipeline Architecture

① Who people are

We use U.S. Census **ACS (American Community Survey)** microdata — **3.5M+ respondents** — vs. Pew ATP (American Trends Panel, ~38K) or GSS (General Social Survey, ~3K/yr) that current models rely on.

② What people believe

Opinion priors come from **79 waves of Pew ATP (2021–2024)** · 7,728 items across 10 domains — health, immigration, economy, environment, technology & more.
Current models filter out demographic groups or cover only one domain.

③ How they respond

Each synthetic agent is prompted with their full demographic profile + calibrated opinion priors, then queried via an LLM to simulate a realistic, intersectional survey response.

04 · THE PRODUCT

05 · RESULTS

~2× `Which 3 facts best predict someone's politics?`

Race, location, and age - not the usual suspects

Standard models default to age, income, and education. We found **race + location + age** explains roughly twice as much opinion variance - at the exact same 3-variable cost.

47% `How much does geography actually matter?`

More than income and education combined

Removing census region alone accounts for a 47% drop in predictive signal. **Where someone lives** is one of the strongest predictors in the dataset.

How much does each variable help predict someone's political opinion?